The Inverse Transform: Recap

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Pseudorandom numbers

A **PRNG** maps a seed to a number and a new seed:

$$\text{seed}_k \longmapsto (U_k, \text{seed}_{k+1})$$

- **Deterministic** (not inherently true randomness)
- The **seed** makes a simulation **reproducible**: `set.seed()`.
- It **repeats** eventually: the LCG after 8, Mersenne-Twister after $2^{19937} - 1$.
- We cannot **prove** a generator is good, only **reject** bad ones.

We will use `runif()` as our PRNG.

The quantile function

For any random variable, with CDF $F_X(t) = \Pr(X \leq t)$:

$$Q_X(p) = \inf\{x : F_X(x) \geq p\}, \qquad p \in [0, 1]$$

- "The **smallest** x whose CDF reaches p."
- Defined for **every** distribution: continuous, discrete, or mixed.
- When F_X is continuous and strictly increasing, $Q_X = F_X^{-1}$.
- In R, q functions: `qnorm`, `qexp`, `qbinom`.

The inverse transform method

For **continuous** X and $U \sim U(0,1)$:

$$Q_X(U) \sim X$$

The recipe:

1. Find the CDF $F(x)$.
2. Set $u = F(x)$ and solve for x. That is $Q(u)$.
3. Return Q applied to `runif(n)`.

Exponential: $u = 1 - e^{-\lambda x}$ gives $Q(u) = -\log(1-u)/\lambda$, so `-log(runif(n)) / rate`.

Corollary (probability integral transform): $F_X(X) \sim U(0,1)$.

Checking a sampler: QQ plots

A quantile function **uniquely determines** a distribution.

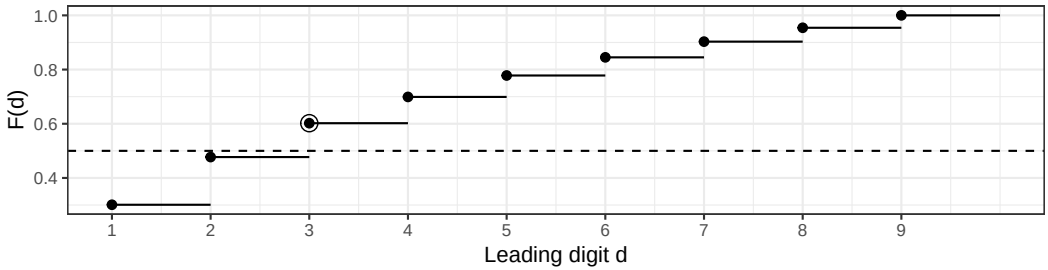
- Plot sample quantiles against the target's quantiles, $Q(u)$ for many u.
- **On the 45 degree line**: the sampler matches the target.
- **Off the line**: it does not.
- `geom_qq(distribution = q_ours)` takes **any** quantile function, including one you wrote.

That last point checks the sampler against **your math**, not against R.

Discrete variables: Q is a step function

The continuous proof fails: there may be no x with $F(x) = u$. The method still works.

$$Q(u) = x \quad \text{where} \quad F(x-1) < u \leq F(x)$$



Benford with $u = 0.5$: $F(2) = 0.477 < 0.5 \leq 0.602 = F(3)$, so $d = 3$.

Two ways to compute a discrete Q

Walk the PMF. Works for any PMF you can compute.

1. Draw u, set $v = 0$.
2. For $x_1 < x_2 < \ldots$: add $p(x_i)$ to v. Stop at the **first** x_i with $v \geq u$.

Closed form. When F has one, solve $F(x-1) < u \leq F(x)$ directly.

- Benford: $F(d) = \log_{10}(d+1)$, so $Q(u) = \lceil 10^u - 1 \rceil$.
- In R: `ceiling(10^runif(n) - 1)`.